

Deep Dive Documentation Map

This folder is the explanation pack for the Smart Microgrid EMS project. It is written for project review, viva, report defense, and operator-facing demonstration.

Files

File	Purpose
01-project-purpose-real-world-value.md	What the project does, real-world usage, and why it improves over rule-based/manual EMS approaches.
02-report-vs-implementation-audit.md	Point-by-point comparison between the submitted report and the implemented project.
03-codebase-formulas-and-runbook.md	Codebase explanation, formulas implemented in code, and how to run the project in one go.
04-dashboard-chart-guide.md	What every dashboard chart/panel means in real time and how a grid operator uses it.
05-real-time-simulation-backend.md	What happens when the live simulation is started, what data is generated, and what can be inferred.
06-outputs-results-and-defense.md	What outputs to present, what generated PNGs show, and how to defend imperfect results honestly.
07-forecasting-explained.md	What is forecasted, why forecasting matters, how results are produced, and where to view them.
08-future-upgrades.md	Future upgrades from the current project state toward a stronger real-world EMS.
09-data-and-eda.md	Data sources, synthetic data, detailed EDA interpretation, and links to EDA artifacts.

Recommended Reading Order

1. Start with [01-project-purpose-real-world-value.md](#).
2. Use [03-codebase-formulas-and-runbook.md](#) to understand the implementation.
3. Use [04-dashboard-chart-guide.md](#) and [05-real-time-simulation-backend.md](#) before a live demo.
4. Use [06-outputs-results-and-defense.md](#) for presentation slides and viva defense.
5. Use [02-report-vs-implementation-audit.md](#) to answer report alignment questions.
6. Use [07-forecasting-explained.md](#), [08-future-upgrades.md](#), and [09-data-and-eda.md](#) for deeper technical questions.